Class activity-10.07.2026

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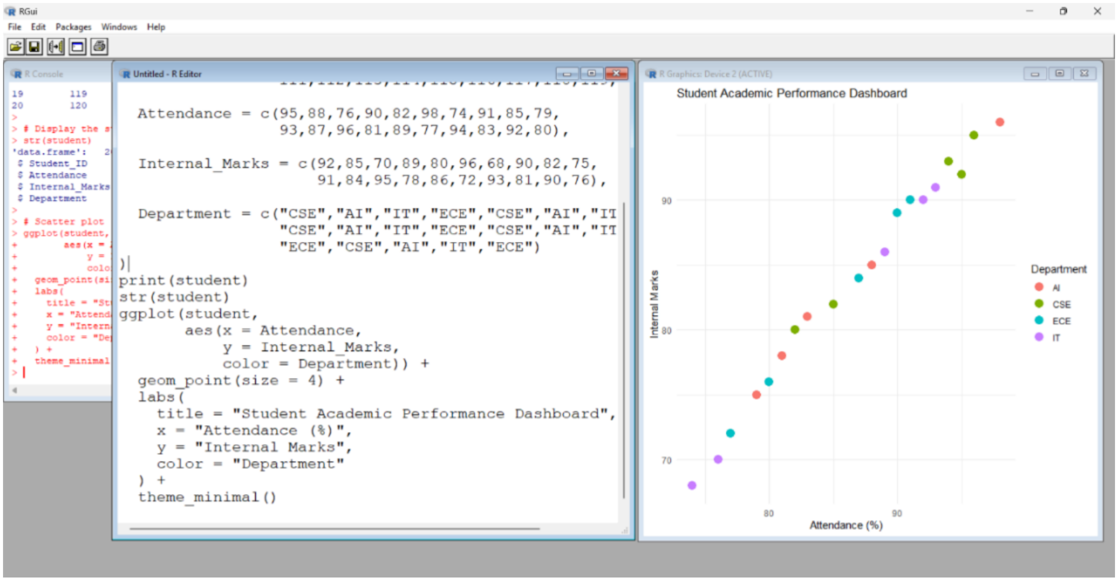
1.Student Performance :

```
install.packages("ggplot2")
library(ggplot2)
student <- data.frame(
Student_ID = c(101,102,103,104,105,106,107,108,109,110,
               111,112,113,114,115,116,117,118,119,120),
  Attendance = c(95,88,76,90,82,98,74,91,85,79,
               93,87,96,81,89,77,94,83,92,80),
Internal_Marks = c(92,85,70,89,80,96,68,90,82,75,
                   91,84,95,78,86,72,93,81,90,76),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE",
               "CSE","AI","IT","ECE","CSE","AI","IT",
               "ECE","CSE","AI","IT","ECE")
)
print(student)
str(student)ggplot(student,
aes(x = Attendance,
         y = Internal_Marks,
color = Department)) +
geom_point(size = 4) +
  labs(
```

```
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
color = "Department"
  ) +
theme_minimal()
```

Output :



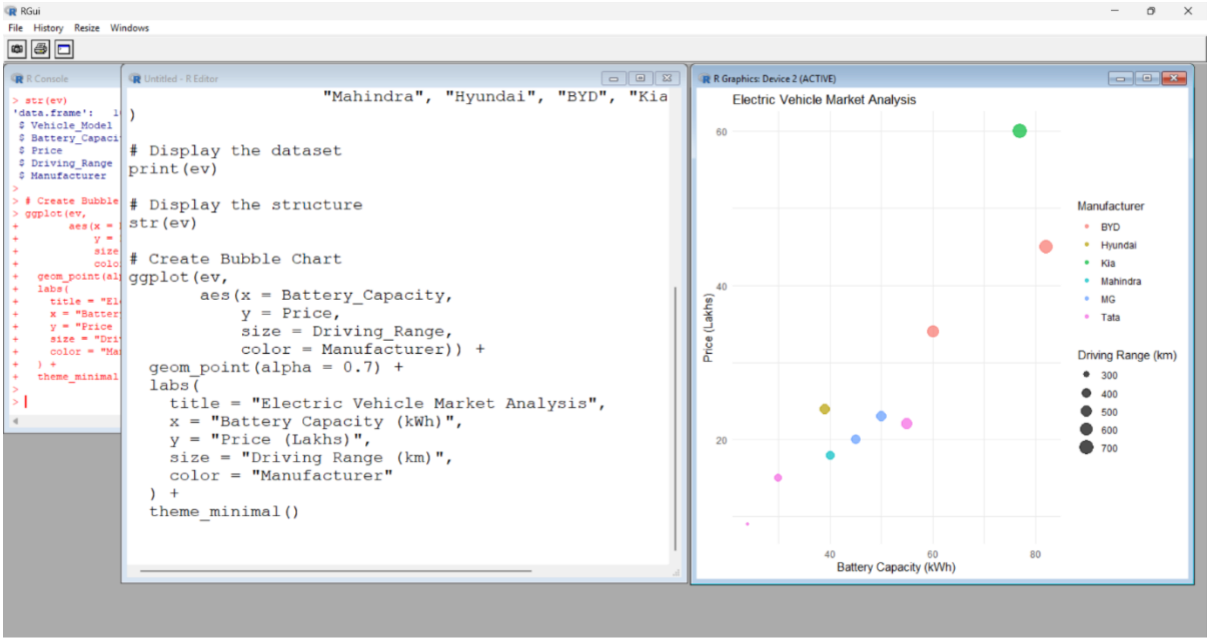
2.Electric Vehicle Market Analysis :

Code:

```
install.packages("ggplot2")
library(ggplot2)
ev<- data.frame(
Vehicle_Model = c("Nexon EV", "Tiago EV", "MG ZS EV", "BYD Atto 3",
                  "Tata Curvv EV", "Mahindra XUV400", "Hyundai Kona",
                  "BYD Seal", "Kia EV6", "MG Windsor EV"),

Battery_Capacity = c(30, 24, 50, 60, 55, 40, 39, 82, 77, 45),
```

```
  Price = c(15, 9, 23, 34, 22, 18, 24, 45, 60, 20),
Driving_Range = c(325, 250, 461, 521, 500, 375, 452, 650, 708, 420),
  Manufacturer = c("Tata", "Tata", "MG", "BYD", "Tata",
                 "Mahindra", "Hyundai", "BYD", "Kia", "MG")
)
print(ev)
str(ev)
ggplot(ev,
aes(x = Battery_Capacity,
          y = Price,
          size = Driving_Range,
color = Manufacturer)) +
geom_point(alpha = 0.7) +
  labs(
    title = "Electric Vehicle Market Analysis",
    x = "Battery Capacity (kWh)",
    y = "Price (Lakhs)",
    size = "Driving Range (km)",
color = "Manufacturer"
  ) +
theme_minimal()
```

Output :



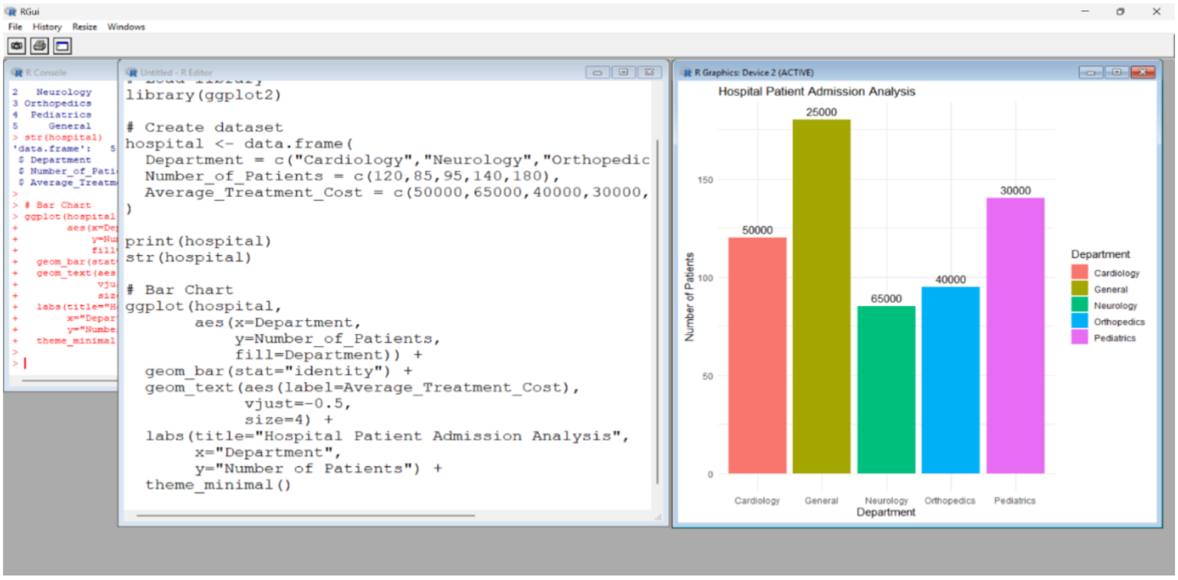
3.Hospital Patient Admission Analysis :

Code :

```
install.packages("ggplot2")

library(ggplot2)

hospital <- data.frame(

  Department =
c("Cardiology","Neurology","Orthopedics","Pediatrics","General"),

Number_of_Patients = c(120,85,95,140,180),

Average_Treatment_Cost = c(50000,65000,40000,30000,25000)

)

print(hospital)

str(hospital)

ggplot(hospital,

aes(x=Department,

          y=Number_of_Patients,
```

```
            fill=Department)) +
geom_bar(stat="identity") +
geom_text(aes(label=Average_Treatment_Cost),
vjust=-0.5,
            size=4) +
  labs(title="Hospital Patient Admission Analysis",
       x="Department",
       y="Number of Patients") +
theme_minimal()
```

Ouput :



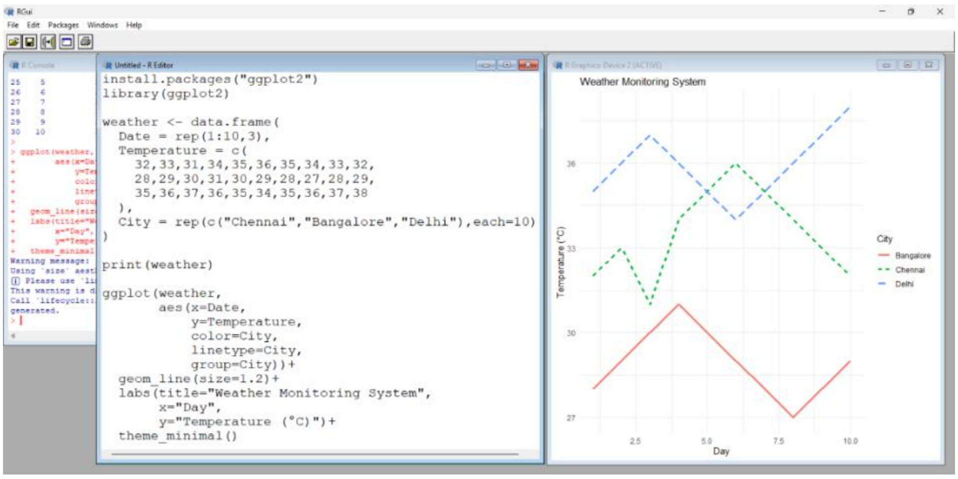
4.Weather Monitoring system

Code :

```
install.packages("ggplot2")
library(ggplot2)
weather <- data.frame(
  Date = rep(1:10,3),
  Temperature = c(
```

```
    32,33,31,34,35,36,35,34,33,32,
    28,29,30,31,30,29,28,27,28,29,
    35,36,37,36,35,34,35,36,37,38
  ),
  City = rep(c("Chennai","Bangalore","Delhi"),each=10)
)
print(weather)
ggplot(weather,
aes(x=Date,
          y=Temperature,
color=City,
linetype=City,
          group=City))+
geom_line(size=1.2)+
  labs(title="Weather Monitoring System",
      x="Day",
      y="Temperature (°C)")+
theme_minimal()
```

Output:

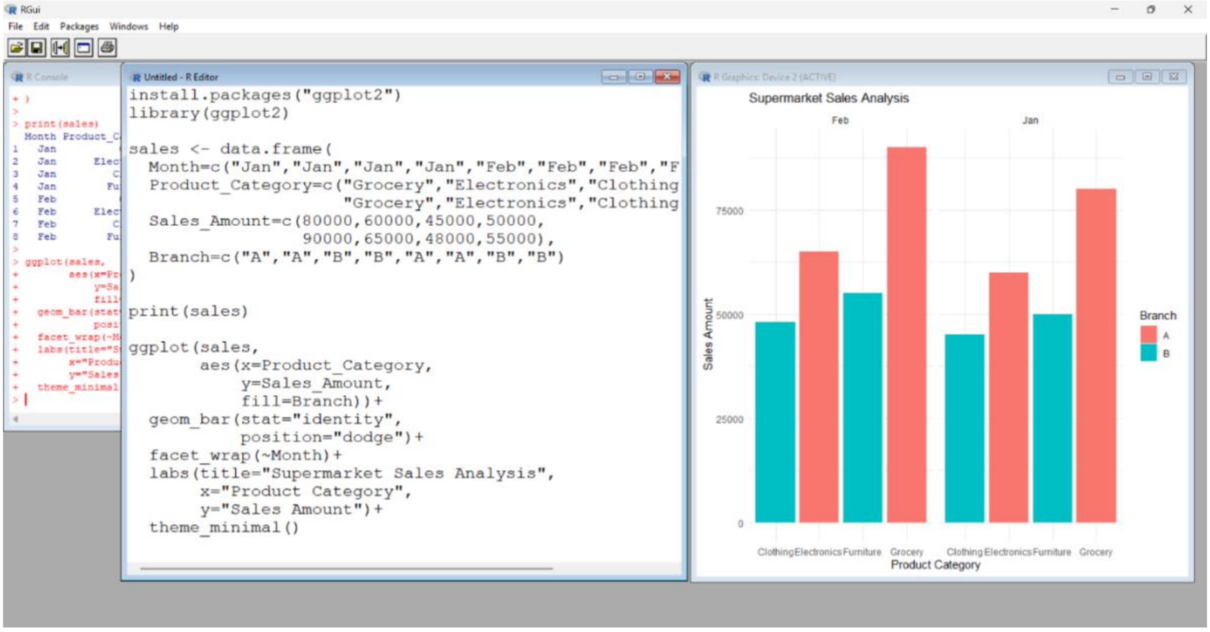


5.Supermarket Sales Analysis :

Code :

```
install.packages("ggplot2")
library(ggplot2)

sales <- data.frame(
  Month=c("Jan","Jan","Jan","Jan","Feb","Feb","Feb","Feb"),
  Product_Category=c("Grocery","Electronics","Clothing","Furniture",
                   "Grocery","Electronics","Clothing","Furniture"),
Sales_Amount=c(80000,60000,45000,50000,
               90000,65000,48000,55000),
  Branch=c("A","A","B","B","A","A","B","B")
)
print(sales)
ggplot(sales,
aes(x=Product_Category,
          y=Sales_Amount,
          fill=Branch))+
geom_bar(stat="identity",
          position="dodge")+
facet_wrap(~Month)+
  labs(title="Supermarket Sales Analysis",
      x="Product Category",
      y="Sales Amount")+
theme_minimal()
```

Output :



6.Employee salary distribution

Code :

```
install.packages("ggplot2")
library(ggplot2)
employee <- data.frame(
Employee_ID=1:20,
  Salary=c(25000,30000,28000,40000,35000,45000,50000,55000,60000,
           65000,70000,72000,75000,80000,85000,90000,95000,
           100000,105000,110000),
  Experience=c(1,2,2,3,3,4,5,5,6,6,7,8,8,9,10,11,12,13,14,15),
  Department=c("HR","HR","IT","IT","Sales","Sales","Finance","Finance",
               "HR","IT","Sales","Finance","HR","IT","Sales",
               "Finance","HR","IT","Sales","Finance")
)
print(employee)
ggplot(employee,aes(x=Salary))+
```

geom_histogram(binwidth=10000,

fill="skyblue",

color="black")+

labs(title="Employee Salary Distribution",

x="Salary",

y="Frequency")+

theme_minimal()

Output :

7.Smart phone analysis

Code :

install.packages("ggplot2")

library(ggplot2)

phone <- data.frame(

Brand=c("Samsung","Apple","OnePlus","Xiaomi","Vivo",

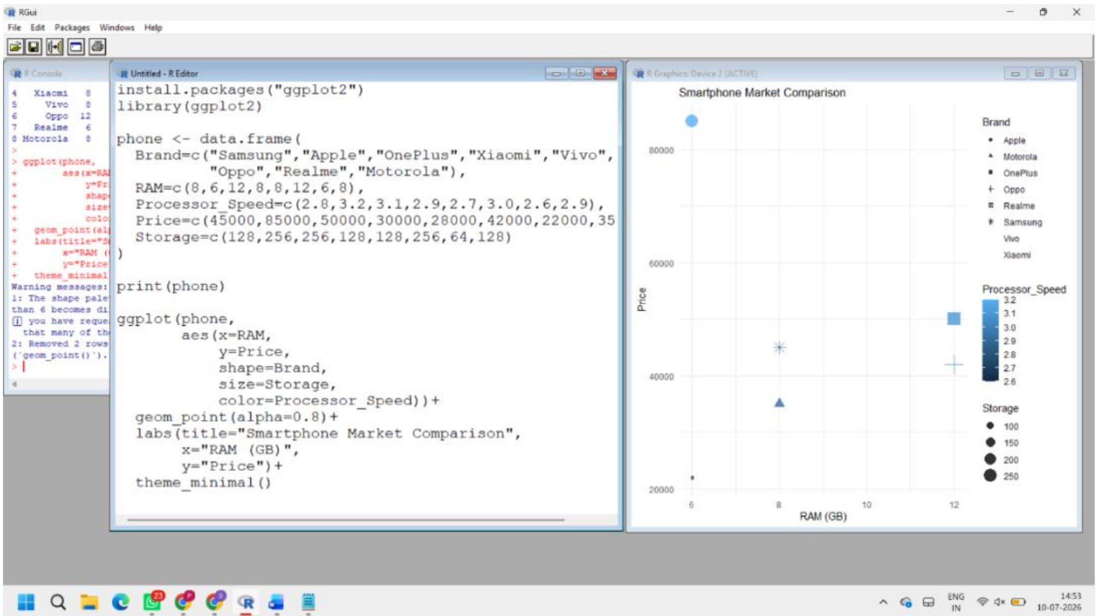
"Oppo","Realme","Motorola"),

RAM=c(8,6,12,8,8,12,6,8),

Processor_Speed=c(2.8,3.2,3.1,2.9,2.7,3.0,2.6,2.9),

```
  Price=c(45000,85000,50000,30000,28000,42000,22000,35000),
  Storage=c(128,256,256,128,128,256,64,128)
)
print(phone)
ggplot(phone,
aes(x=RAM,
           y=Price,
           shape=Brand,
           size=Storage,
color=Processor_Speed))+
geom_point(alpha=0.8)+
  labs(title="Smartphone Market Comparison",
       x="RAM (GB)",
       y="Price")+
theme_minimal()
```

Output :



8.Online Movie Rating Analysis

Code :install.packages("ggplot2")

```
library(ggplot2)

movies <- data.frame(
  Genre=c("Action","Comedy","Drama","Horror","Sci-Fi",
          "Action","Comedy","Drama","Horror","Sci-Fi"),
Release_Year=c(2023,2023,2023,2023,2023,
               2024,2024,2024,2024,2024),
Average_Rating=c(8.5,7.2,8.8,6.9,9.0,
                 8.7,7.5,9.1,7.0,9.3),
Number_of_Reviews=c(1500,1200,1800,900,2000,
                    1600,1300,1900,950,2200)
)

print(movies)

ggplot(movies,
aes(x=Release_Year,
          y=Genre,
          fill=Average_Rating))+
geom_tile(color="white")+
scale_fill_gradient(low="yellow",high="red")+
  labs(title="Movie Rating Heatmap",
       x="Release Year",
       y="Genre")+ theme_minimal()
```

Output :

9. COVID-19 State-wise Analysis

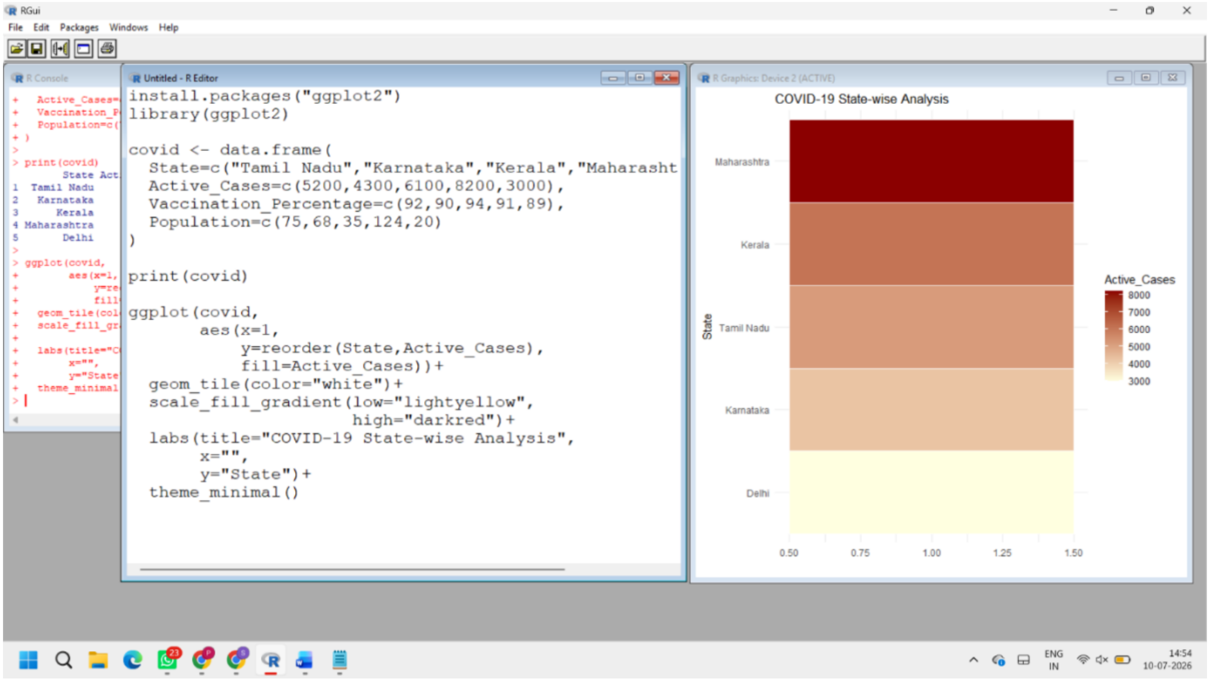
CODE: install.packages("ggplot2")

```
library(ggplot2)


covid <- data.frame(
  State=c("Tamil Nadu","Karnataka","Kerala","Maharashtra","Delhi"),
Active_Cases=c(5200,4300,6100,8200,3000),
Vaccination_Percentage=c(92,90,94,91,89),
  Population=c(75,68,35,124,20)
)
print(covid)
ggplot(covid,
aes(x=1,
        y=reorder(State,Active_Cases),
        fill=Active_Cases))+
geom_tile(color="white")+
scale_fill_gradient(low="lightyellow",
                high="darkred")+
  labs(title="COVID-19 State-wise Analysis",
     x="",
```

y="State")+

theme_minimal()

OUTPUT:



10. Iris Flower Classification

CODE : install.packages("ggplot2")

library(ggplot2)

print(head(iris))

ggplot(iris,

aes(x=Sepal.Length,

    y=Petal.Length,

color=Species,

    shape=Species,

    size=Sepal.Width))+

geom_point(alpha=0.8)+

  labs(title="Iris Flower Classification",

    x="Sepal Length",

```
       y="Petal Length")+
theme_minimal()
```

OUTPUT :